Stationary measures of colored interacting particle systems

Matthew Nicoletti

UC Berkeley

2024

(joint work with A. Aggarwal and L. Petrov)

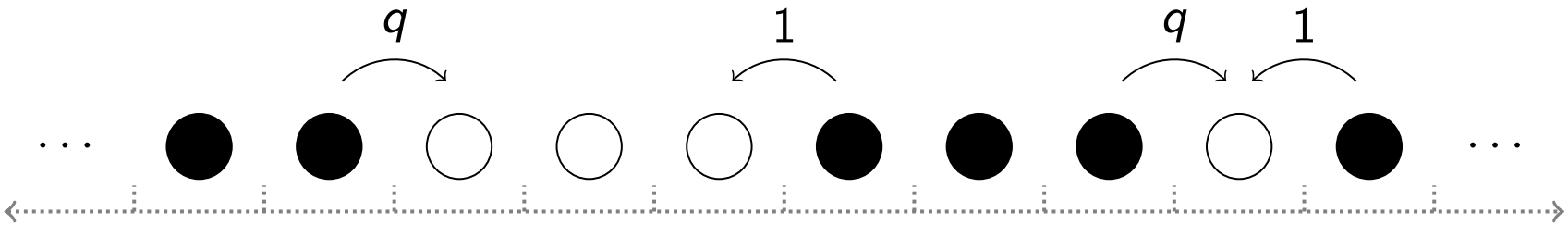
Outline

1. Single species and multi-species interacting particle systems
2. Stationary measures
3. $U_q(\hat{sl}_n)$ vertex models
4. Queue vertex model and stationary measures
5. Stationarity in the quadrant: Burke's property from Yang–Baxter and colored analogs
6. Future directions

Single species and multi-species interacting particle systems

ASEP: one color case

1. ASEP is a continuous time Markov process. States are configurations of particles on sites of a lattice (the line $\mathbb{Z}$ or the ring $\mathbb{Z}/N\mathbb{Z}$).
2. Each particle jumps to the right at rate q and to the left at rate 1; if another particle occupies the target site, the jump is blocked.



mASEP: The multicolored ASEP

1. The particles have *species* or *colors* in $\{1, \ldots, n\}$.
2. Particles of colors (i_k, i_{k+1}) at adjacent sites $k, k+1$ swap (leading to a local configuration (i_{k+1}, i_k)) with rate

$$\mathrm{Rate}((i_k, i_{k+1}) \to (i_{k+1}, i_k)) = \begin{cases} q, & i_k > i_{k+1}; \\ 1, & i_k < i_{k+1}, \end{cases} \tag{1}$$

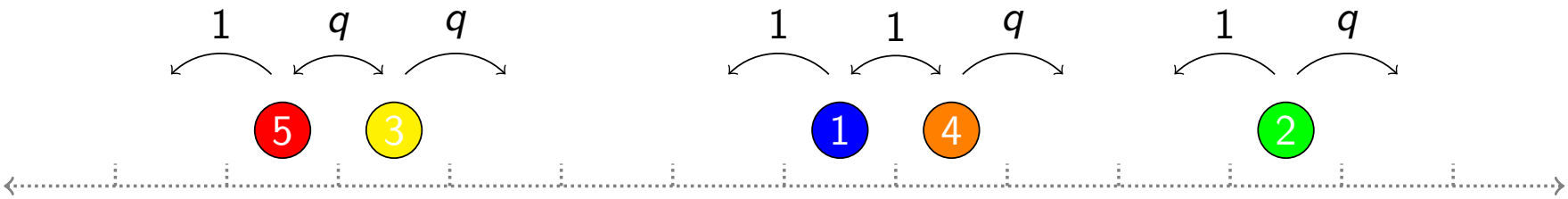
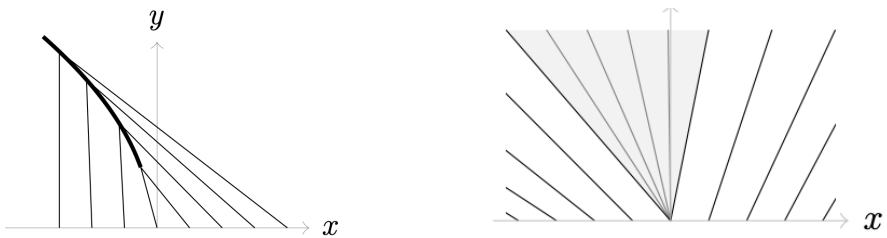


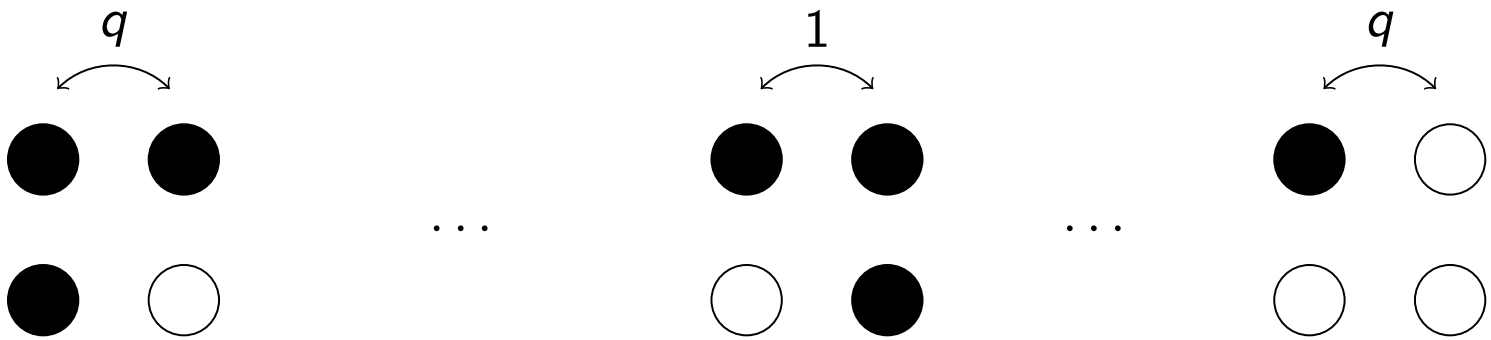
Figure from [BW18]

mASEP: Motivation

1. Hydrodynamics. A single type 1 (weaker) particle amidst many type 2 particles will follow characteristics. They describe "microscopic structure of characteristics":



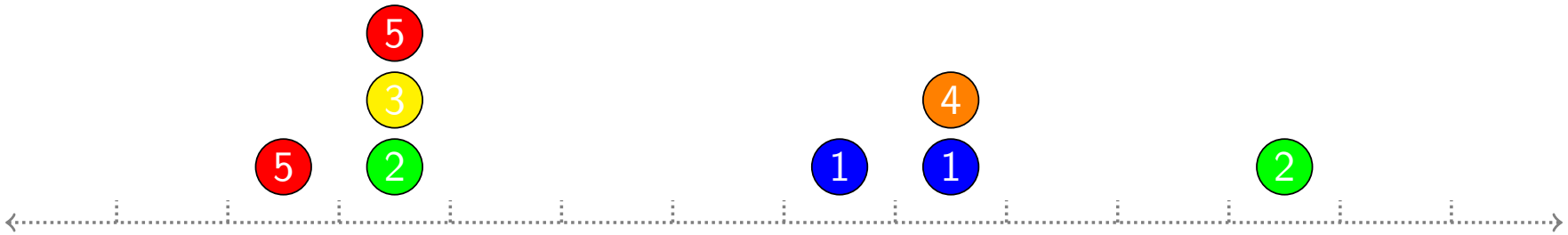
2. "Attractive couplings". Discrepancies evolve as second class particles:



3. Algebraic and combinatorial structure. Stationary measures are related to multiline queues, nonsymmetric Macdonald polynomials, etc...

Multispecies q-Boson and q-PushTASEP

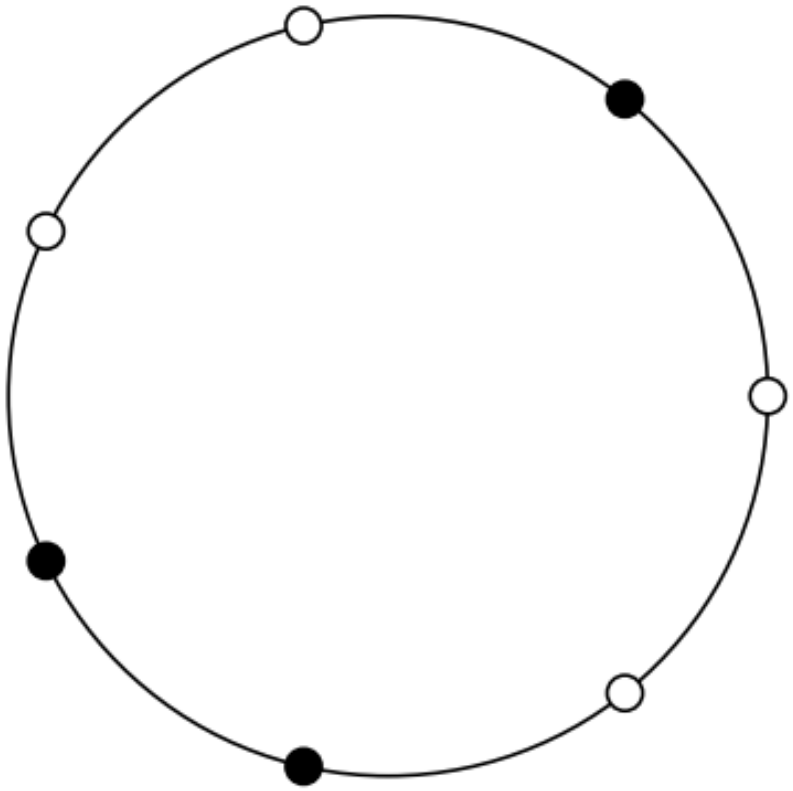
1. In these two particle systems, multiple particles (of any color) can occupy one site.
2. The q-Boson process is *zero range*: Transition rates from k to $k+1$ depend only on the contents at site k.
3. The q-PushTASEP is a q-deformation of a variant of TASEP with *pushing*: Jumps of particles can activate motion of other particles.



Stationary measures

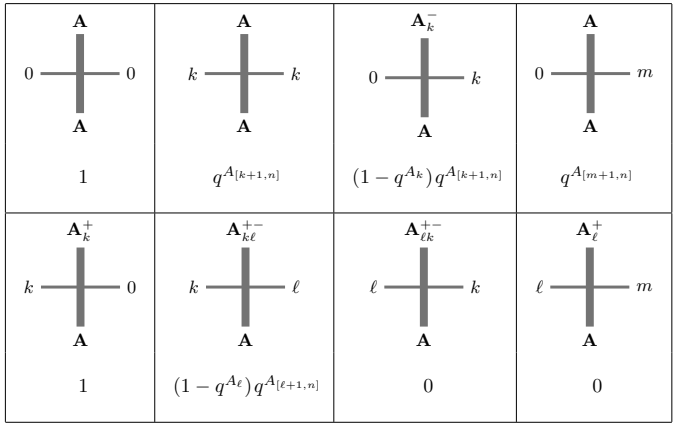
ASEP warmup

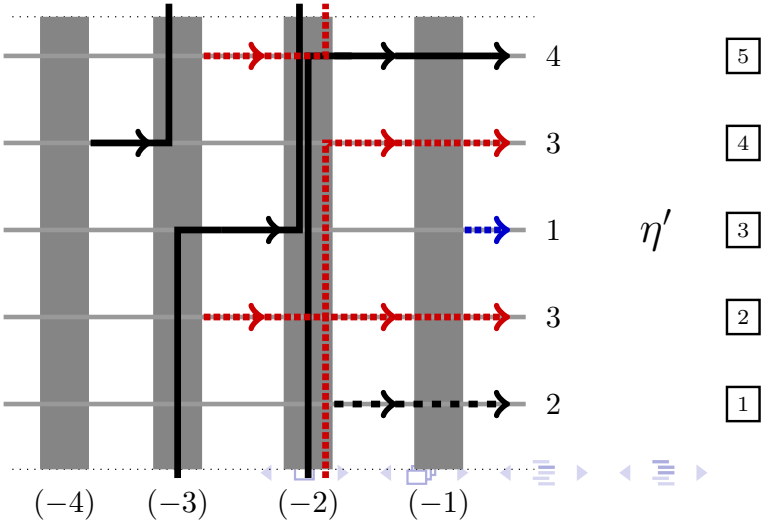
1. Consider the sector $\{\eta\}$ with $N_1 > 0$ particles on the ring of size N.
2. One can check that the uniform distribution (over all $\binom{N}{N_1}$ such configurations) is stationary under ASEP.



mASEP Stationary measures: Multiline queues

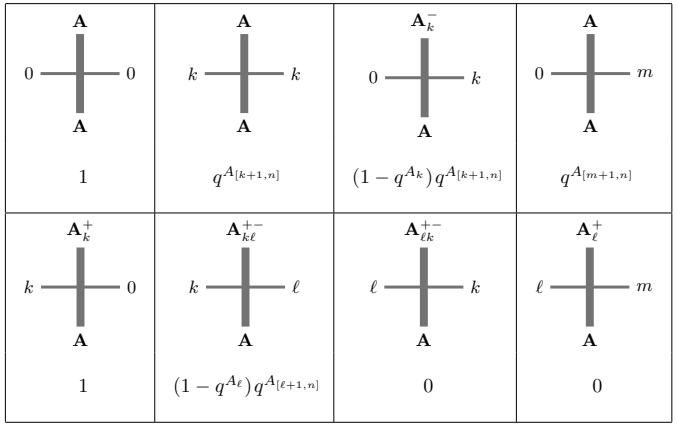
1. Martin [Mar20] realizes stationary probabilities using *multiline queues*. There are n columns on a cylinder. Queue configurations are path configurations. In column $-m$, paths of color m can be created. Paths may wind around the cylinder.

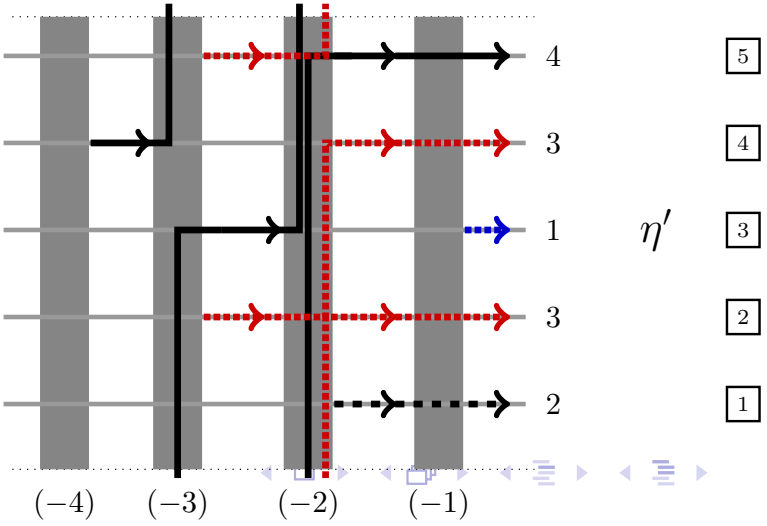




mASEP Stationary measures: Multiline queues

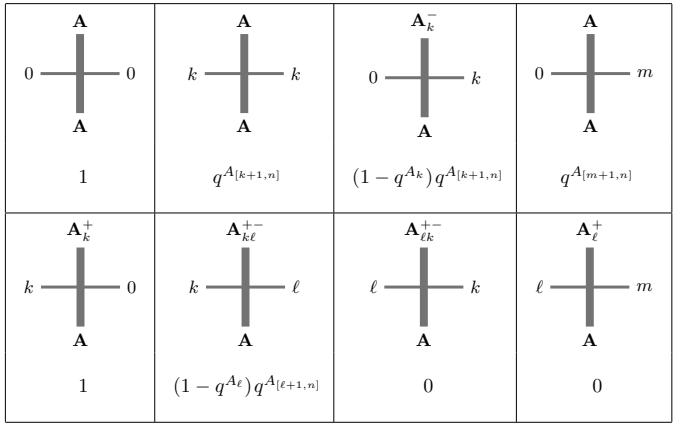
1. Martin [Mar20] realizes stationary probabilities using *multiline queues*. There are n columns on a cylinder. Queue configurations are path configurations. In column $-m$, paths of color m can be created. Paths may wind around the cylinder.
2. Each allowed queue configuration has a *weight*, the product of local weights. Summing weights over states with fixed boundary condition η' gives the stationary weight of η'.

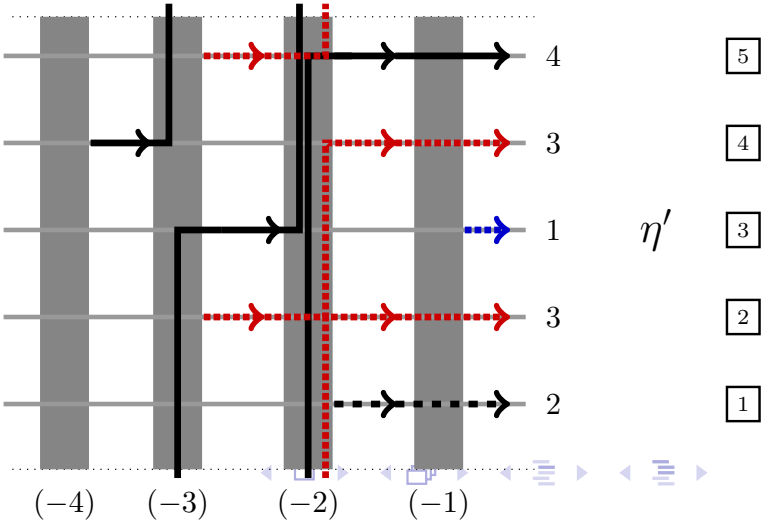




mASEP Stationary measures: Multiline queues

1. Martin [Mar20] realizes stationary probabilities using *multiline queues*. There are n columns on a cylinder. Queue configurations are path configurations. In column $-m$, paths of color m can be created. Paths may wind around the cylinder.
2. Each allowed queue configuration has a *weight*, the product of local weights. Summing weights over states with fixed boundary condition η' gives the stationary weight of η'.
3. Weights of configurations match terms in a *matrix product state* $P(\eta) = \frac{1}{Z}\mathrm{Tr}\left(\prod_{j=1}^{N} X_j\right)$ known to be stationary [PEM09].





The main idea

1. Our goal: Construct stationary measures, or leading eigenvectors of the transition matrices, from matrix elements of a "transfer matrix" $\mathfrak{Q}$ of a *quantum integrable vertex model*.

The main idea

1. Our goal: Construct stationary measures, or leading eigenvectors of the transition matrices, from matrix elements of a "transfer matrix" $\mathfrak{Q}$ of a *quantum integrable vertex model*.
2. We consider probabilities proportional to $\mathrm{wt}(\mathbf{V}) := \langle \emptyset | \mathfrak{Q} | \mathbf{V} \rangle > 0$, where $\emptyset$ is the state with no particles.

The main idea

1. Our goal: Construct stationary measures, or leading eigenvectors of the transition matrices, from matrix elements of a "transfer matrix" $\mathfrak{Q}$ of a *quantum integrable vertex model*.
2. We consider probabilities proportional to $\mathrm{wt}(\mathbf{V}) := \langle \emptyset | \mathfrak{Q} | \mathbf{V} \rangle > 0$, where $\emptyset$ is the state with no particles.
3. Then, if the dynamics of the particle system can be (approximately) generated by a transfer matrix T in the same commuting family, then

$$\langle \emptyset | \mathfrak{Q}\, T | \mathbf{V} \rangle = \langle \emptyset | T \mathfrak{Q} | \mathbf{V} \rangle = \langle \emptyset | \mathfrak{Q} | \mathbf{V} \rangle$$

since $\langle \emptyset | T = \langle \emptyset |$.

The main idea

1. Our goal: Construct stationary measures, or leading eigenvectors of the transition matrices, from matrix elements of a "transfer matrix" $\mathfrak{Q}$ of a *quantum integrable vertex model*.
2. We consider probabilities proportional to $\mathrm{wt}(\mathbf{V}) := \langle \emptyset | \mathfrak{Q} | \mathbf{V} \rangle > 0$, where $\emptyset$ is the state with no particles.
3. Then, if the dynamics of the particle system can be (approximately) generated by a transfer matrix T in the same commuting family, then

$$\langle \emptyset | \mathfrak{Q}\, T | \mathbf{V} \rangle = \langle \emptyset | T \mathfrak{Q} | \mathbf{V} \rangle = \langle \emptyset | \mathfrak{Q} | \mathbf{V} \rangle$$

   since $\langle \emptyset | T = \langle \emptyset |$.
4. The main challenge is constructing a vertex model $\mathfrak{Q}$ that can "create particles" from the empty state $\emptyset$, which contains no particles.

Main theorem

We construct a unified family of **vertex model partition functions** which compute stationary measures:

Theorem (Aggarwal, N., Petrov)

For each model (mASEP, q-Boson, or q-PushTASEP), by choosing appropriate parameters, we have

$$\mathrm{Prob}_{N_1,\ldots,N_n}(\mathbf{V}) = \frac{1}{Z}\langle\emptyset|\mathfrak{Q}|\mathbf{V}\rangle.$$

Further degenerating leftover parameters, we are able to match to

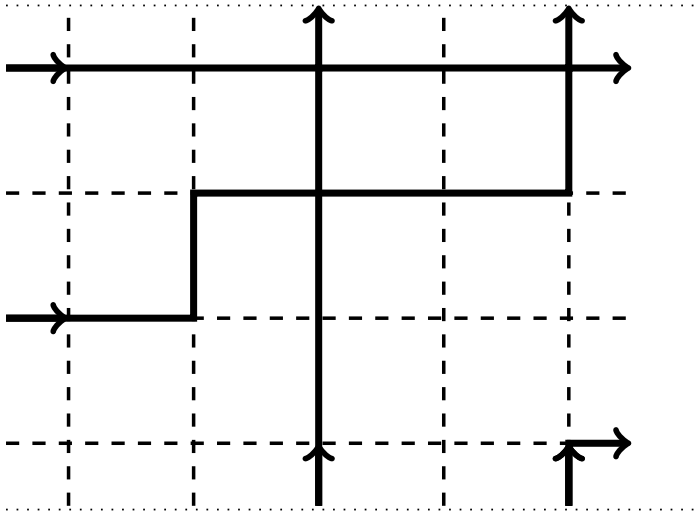
1. known mASEP multiline queues and matrix product states [Martin'20] [Evans-Mallick-Prolhac'09].
2. a known q-Boson multiline queue stationary measure [Ayyer-Mandelshtam-Martin'22].

We also obtain new stationary measures for q-PushTASEP.

$U_q(\hat{sl}_n)$ vertex models

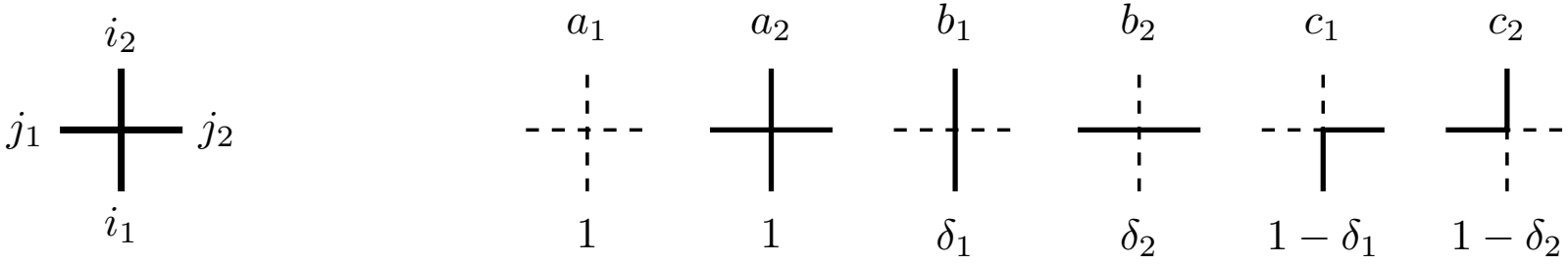
Six-vertex model: The one color case

1. *Path or arrow configurations* are ensembles of up-right paths flowing through a lattice; we focus on a cylinder.
2. Each vertex has a *weight*. The weight assigns a complex number to each local configuration.
3. The total weight of a configuration S is the product of vertex weights: $w(S) = \prod_{i,j} w(i,j;S)$. The *partition function* is $\sum_S w(S)$, where the sum is over configurations satisfying some *boundary condition*.



Six-vertex model: The one color case

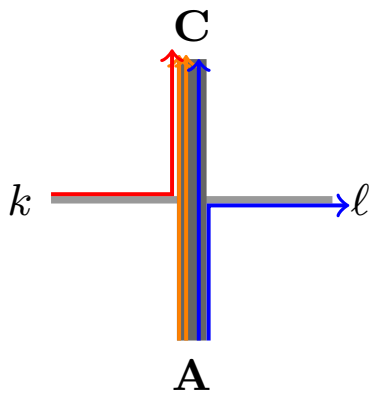
1. By *path conservation* or *arrow conservation* there are six possible local configurations (hence *six-vertex model*).
2. Each vertex (local configuration) has a *weight* $w(i_1, j_1; i_2, j_2)$.

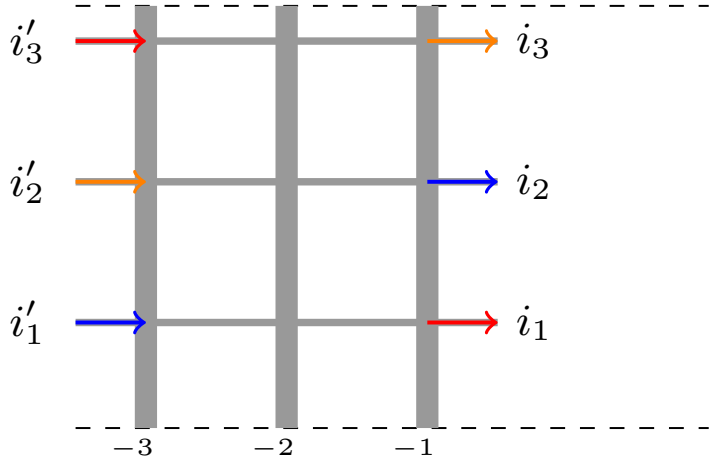


3. The weights with $a_1 = a_2 = 1$, $b_1 = \delta_1$, $c_1 = 1 - \delta_1$, $b_2 = \delta_2$, $c_2 = 1 - \delta_2$ are *stochastic*: $\sum_{i_2, j_2} w(i_1, j_1; i_2, j_2) = 1$.

Higher rank, higher spin partition functions

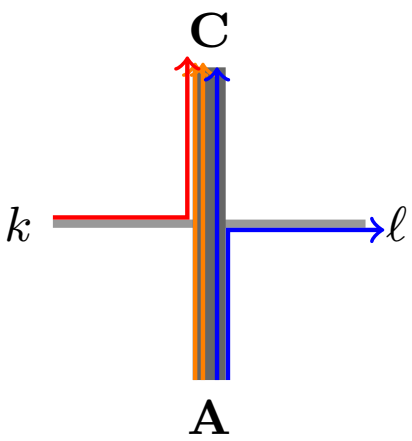
1. *Higher rank* means paths can have n different *colors* or *types*. *Higher spin* means that any number of paths may occupy vertical edges.
2. We denote local arrow configurations in higher spin edges by $\mathbf{A} = (A_1, \ldots, A_n), \mathbf{C} = (C_1, \ldots, C_n) \in \mathbb{Z}^n_{\geq 0}$. Here A_i denotes the number of arrows of color i. Again we have the *arrow conservation property*: $\mathbf{A} + \mathbf{1}_{k>0}\mathbf{e}_k = \mathbf{C} + \mathbf{1}_{\ell>0}\mathbf{e}_\ell$.

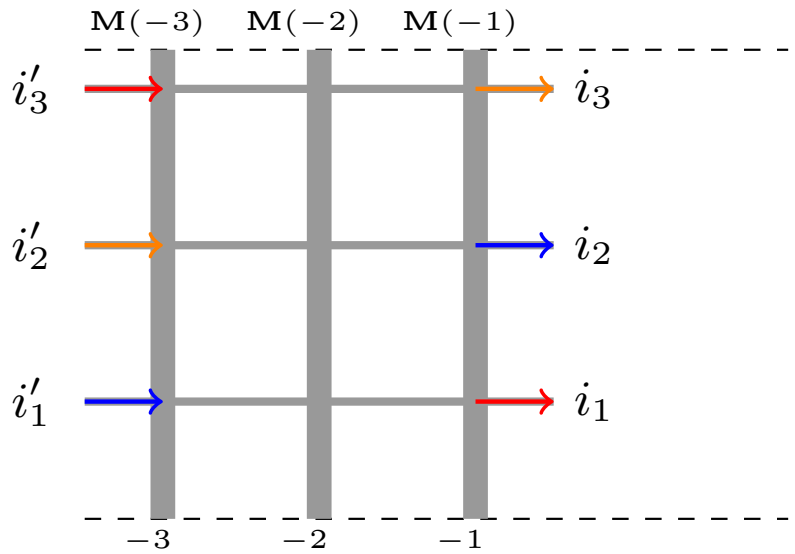




Higher rank, higher spin partition functions

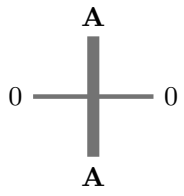
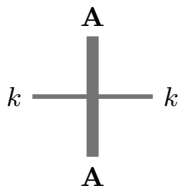
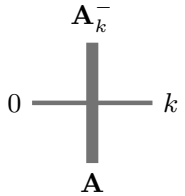
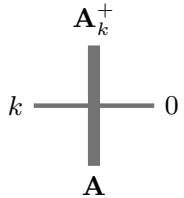
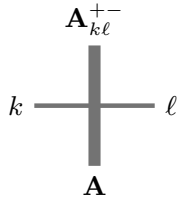
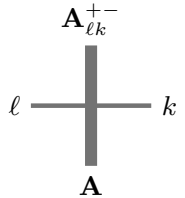
1. To compute the *partition function* we sum over all allowed configurations. The weight of a configuration is the product of local vertex weights.
2. In particular, we sum over all windings $\mathbf{M}(-1), \mathbf{M}(-2), \ldots \in \mathbb{Z}^n_{\geq 0}$.





Higher spin stochastic L weights

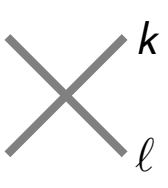
1. Stochastic L weights are parameterized by a *spin parameter* s, and a *spectral parameter* x.
2. Denote $L_{s,x}(\mathbf{A}, k; \mathbf{B}, \ell)$ as the weight of the local arrow configuration specified by $\mathbf{A}, \mathbf{B} \in \mathbb{Z}^n_{\geq 0}$, $k, \ell \in \{0, 1, \ldots, n\}$ as:

 $\frac{1 - sxq^{\lvert\mathbf{A}\rvert}}{1 - sx}$	 $\frac{(-sx + s^2 q^{A_k}) q^{\mathbf{A}_{[k+1,n]}}}{1 - sx}$	 $\frac{-sx(1 - q^{A_k}) q^{\mathbf{A}_{[k+1,n]}}}{1 - sx}$
 $\frac{1 - s^2 q^{\lvert\mathbf{A}\rvert}}{1 - sx}$	 $\frac{-sx(1 - q^{A_\ell}) q^{\mathbf{A}_{[\ell+1,n]}}}{1 - sx}$	 $\frac{-s^2(1 - q^{A_k}) q^{\mathbf{A}_{[k+1,n]}}}{1 - sx}$

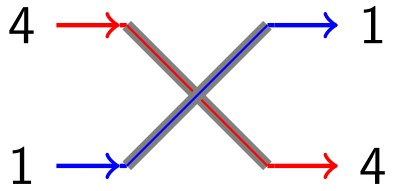
3. We have $\sum_{\mathbf{B},\ell} L_{s,x}(\mathbf{A}, k; \mathbf{B}, \ell) = 1$.

Spin $\frac{1}{2}$: The R-matrix and Yang–Baxter equation

1. Setting $s = q^{-\frac{1}{2}}$, we obtain "cross weights". At most one arrow can occupy each edge. Define the R-matrix [Jim86, Baz85]

$$R_z(i,j;k,\ell) := w\Bigg($$  $$\Bigg)$$

e.g. $R_z(1,4;1,4) = \frac{1-z}{1-qz}$



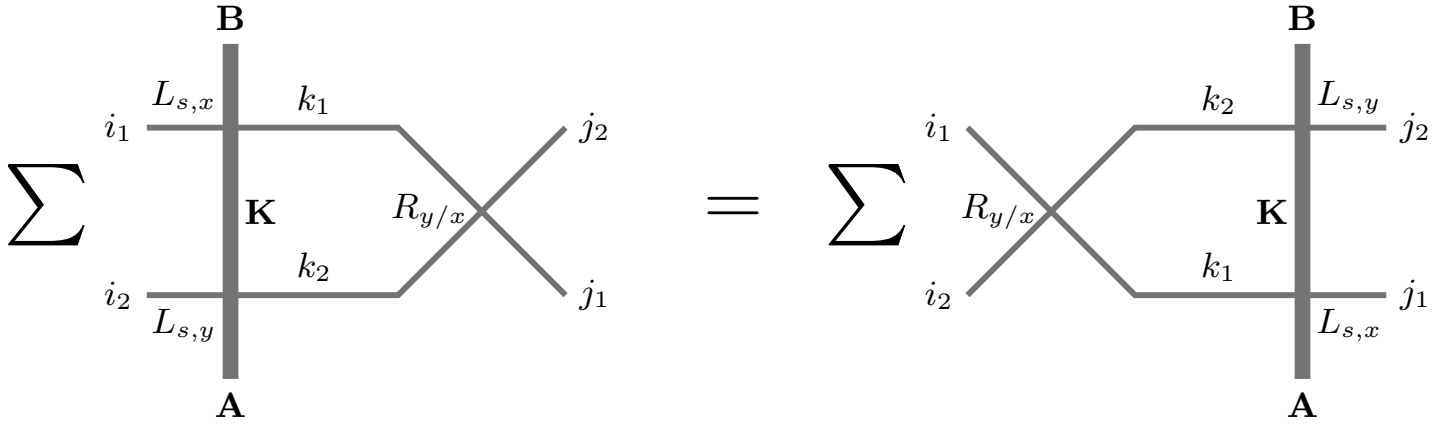
as follows: For $i,j \in \{0,1,\ldots,n\}$

$$R_z(i,i;i,i) = 1, \quad i \in \{0,1,\ldots,n\},$$

$$\left.\begin{aligned} R_z(j,i;j,i) &= \frac{q(1-z)}{1-qz}, & R_z(i,j;i,j) &= \frac{1-z}{1-qz} \\ R_z(j,i;i,j) &= \frac{1-q}{1-qz}, & R_z(i,j;j,i) &= \frac{(1-q)z}{1-qz} \end{aligned}\right\} \quad i<j. \tag{2}$$

R-matrix and Yang–Baxter equation

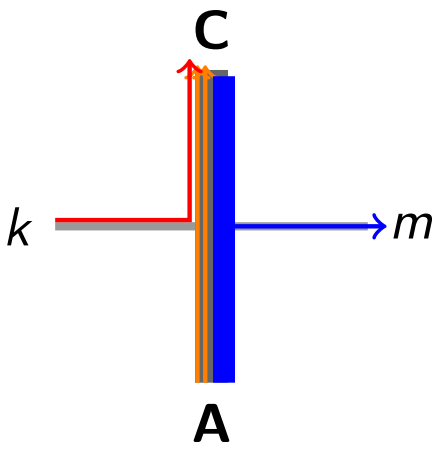
1. The *Yang–Baxter equation* is the equality of partition functions on two different three-vertex graphs.
2. For any fixed $i_1, i_2, j_1, j_2 \in \{0, 1, \ldots, n\}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{Z}^n_{\geq 0}$, we have



The sums in both sides are taken over all $k_1, k_2 \in \{0, 1, \ldots, n\}$ and $\mathbf{K} \in \mathbb{Z}^n_{\geq 0}$.

Digression: A trick to be able to "create paths"

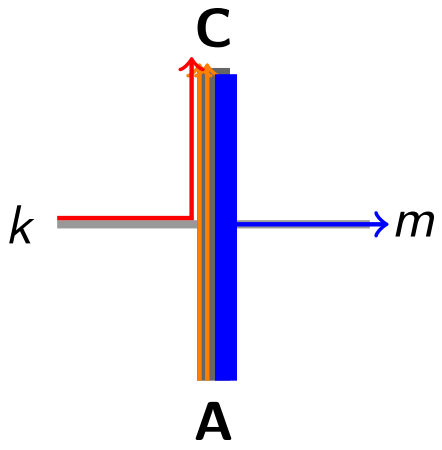
1. Let $m \in \{1, \ldots, n\}$. We define a new vertex weight $\mathbb{W}_{s,x}^{(-m)}$.
2. Let us take the limit of the vertex weight $L_{s,x}(\mathbf{A}, k; \mathbf{C}, \ell)$, sending $A_m, C_m \to +\infty$, with $C_m - A_m \in \{-1, 0, 1\}$ fixed. Denote the result by $\mathbb{W}_{s,x}^{(-m)}(\mathbf{A}, k; \mathbf{C}, \ell)$. Type m paths can now be "created" with weights $\mathbb{W}_{s,x}^{(-m)}$.
3. The Yang–Baxter Equation is preserved!

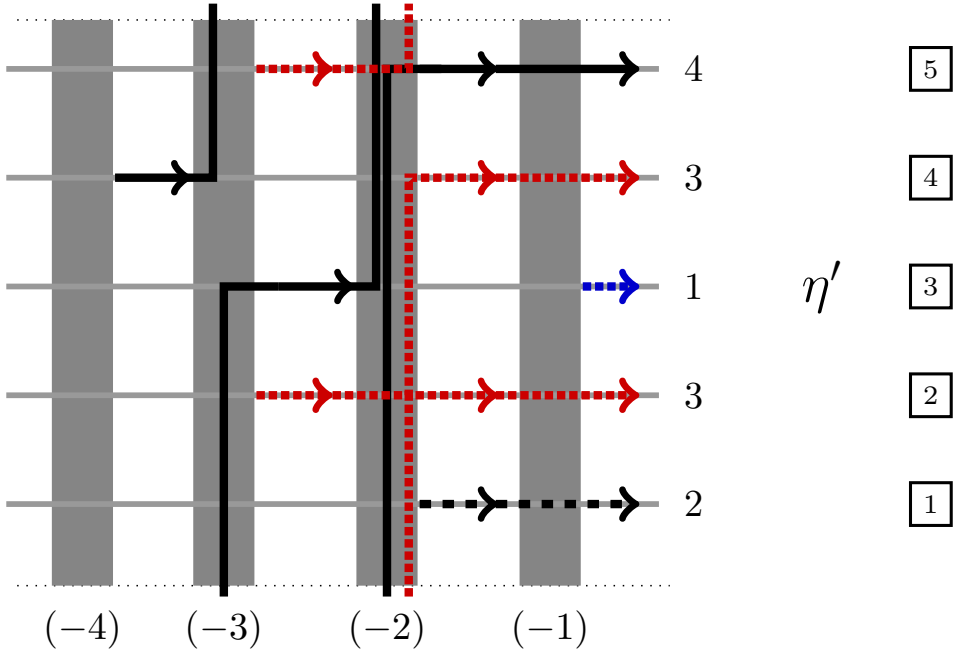


$\mathbf{A}$, 0, 0, $\mathbf{A}$: $\dfrac{1}{1-sx}$	$\mathbf{A}$, k, k, $\mathbf{A}$: $\dfrac{(x - sq^{A_k})\, q^{\mathbf{A}_{[k+1,n]}}}{1-sx}$	$\mathbf{A}_k^-$, 0, k, $\mathbf{A}$: $\dfrac{x\,(1-q^{A_k})\, q^{\mathbf{A}_{[k+1,n]}}}{1-sx}$	$\mathbf{A}$, 0, m, $\mathbf{A}$: $\dfrac{x\, q^{\mathbf{A}_{[m+1,n]}}}{1-sx}$
$\mathbf{A}_k^+$, k, 0, $\mathbf{A}$: $\dfrac{1}{1-sx}$	$\mathbf{A}_{k\ell}^{+-}$, k, ℓ, $\mathbf{A}$: $\dfrac{x\,(1-q^{A_\ell})\, q^{\mathbf{A}_{[\ell+1,n]}}}{1-sx}$	$\mathbf{A}_{\ell k}^{+-}$, ℓ, k, $\mathbf{A}$: $\dfrac{s\,(1-q^{A_k})\, q^{\mathbf{A}_{[k+1,n]}}}{1-sx}$	$\mathbf{A}_\ell^+$, ℓ, m, $\mathbf{A}$: $\dfrac{s\, q^{\mathbf{A}_{[m+1,n]}}}{1-sx}$

Digression: A trick to be able to "create paths"

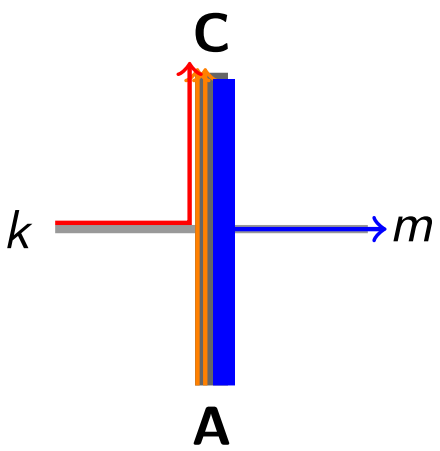
1. If we place these weights $\mathbb{W}_{s,x}^{(-m)}$ at vertices $(-m, j)$ on a cylinder $\{-n, \ldots, -1\} \times \mathbb{Z}/N\mathbb{Z}$, we can *create paths* from the empty state. We call this the *queue specialized vertex model*.
2. In column $-m$, paths of type m can be created, and we sum over windings of colors $> m$. Denote $\langle \emptyset | \mathfrak{Q} | \eta' \rangle$ as the resulting partition function.
3. Combinatorially, this is the same as Martin's queue.





Matching to Martin's queue

1. If we choose parameters $s = 0$, $x = 1$ in the table from two slides ago, we *exactly* match partition functions $\langle \emptyset | \mathfrak{Q} | \eta' \rangle$ to Martin's multiline queue weights.
2. Below we reproduce the weights in column $-m$.



$\mathbf{A}$, 0, 0, $\mathbf{A}$: 1	$\mathbf{A}$, k, k, $\mathbf{A}$: $q^{A_{[k+1,n]}}$	$\mathbf{A}_k^-$, 0, k, $\mathbf{A}$: $(1-q^{A_k})\,q^{A_{[k+1,n]}}$	$\mathbf{A}$, 0, m, $\mathbf{A}$: $q^{A_{[m+1,n]}}$
$\mathbf{A}_k^+$, k, 0, $\mathbf{A}$: 1	$\mathbf{A}_{k\ell}^{+-}$, k, ℓ, $\mathbf{A}$: $(1-q^{A_\ell})\,q^{A_{[\ell+1,n]}}$	$\mathbf{A}_{\ell k}^{+-}$, ℓ, k, $\mathbf{A}$: 0	$\mathbf{A}_\ell^+$, ℓ, m, $\mathbf{A}$: 0

Matching to Martin's queue

1. We obtain:

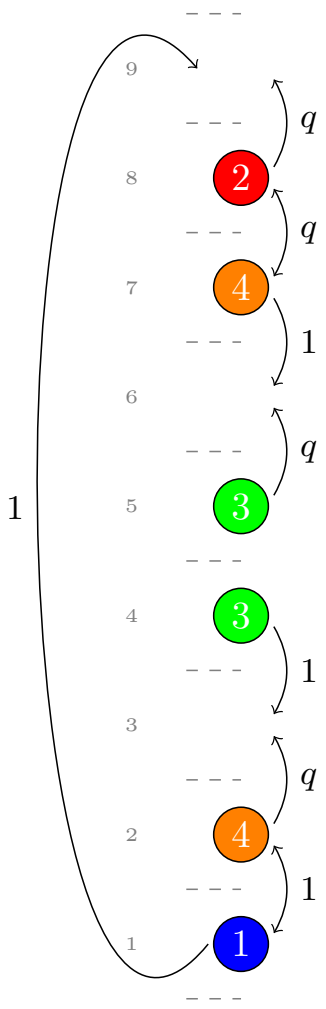
Proposition (Aggarwal, N., Petrov)

Martin's queue model is a special case of the queue specialized vertex model.

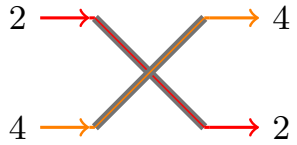
2. Related works include Ferrari-Martin [FM07], Prolhac-Evans-Mallick [PEM09] (Matrix product stationary state), Cantini-de Gier-Wheeler [CdGW15] (Matrix product formula for Macdonald polynomials), Corteel-Mandelshtam-Williams [CMW18] (multiline queue formulas for nonsymmetric Macdonalds), Martin [Mar20] (multiline queue, and sampling algorithm), Ayyer-Mandelshtam-Martin [AMM22] (queues for q-Boson stationary measure), Corteel-Keating [CK23].
3. Our goal is to unify many of the approaches above, and to provide an alternative direct proof of stationarity using vertex model machinery, i.e. the Yang–Baxter equation.

The mASEP dynamics from stochastic R-matrix

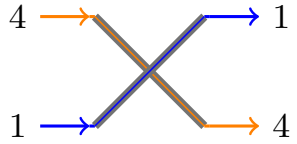
Entries of the stochastic matrix $R_{1-\epsilon}$ approximates jump rates:



$(RP)_{42,24} = R(4,2;4,2) = \frac{1}{1-q} q\epsilon$

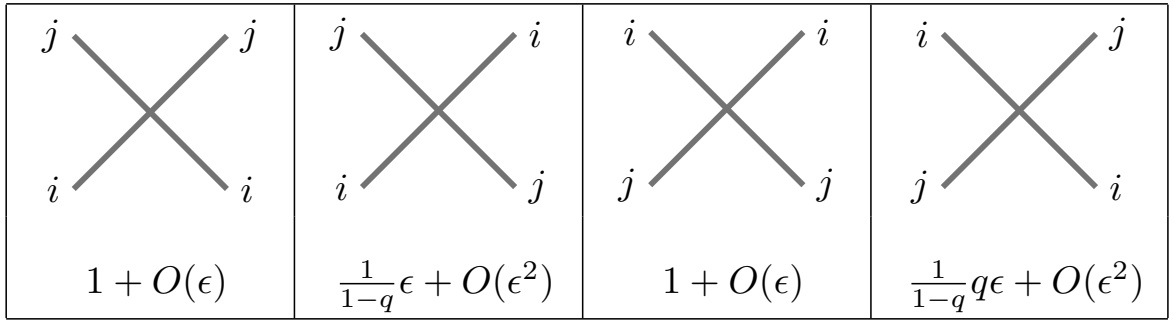


$(RP)_{14,41} = R(1,4;1,4) = \frac{1}{1-q} \epsilon$

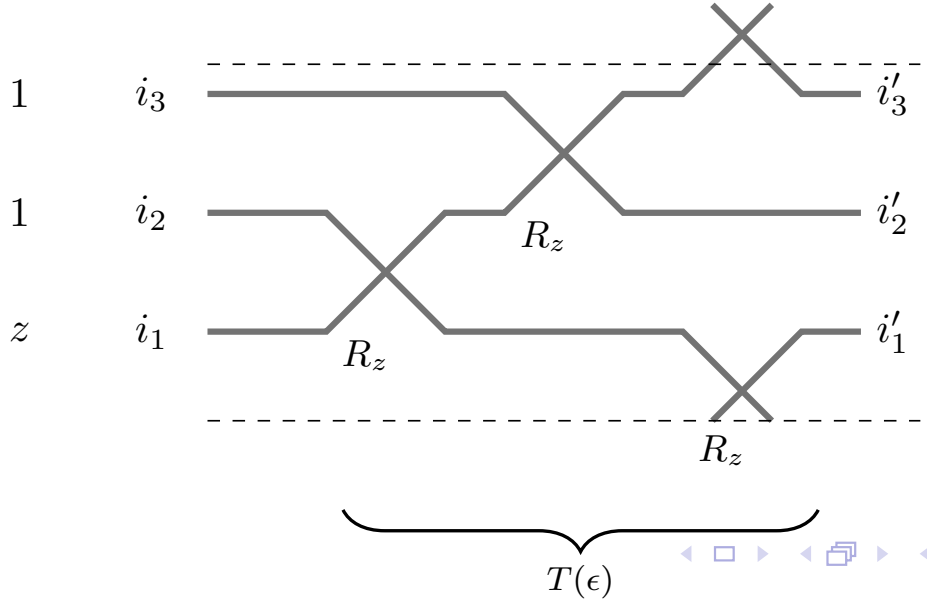


The mASEP dynamics from stochastic R-matrix

1. Let $z = 1 - \epsilon$. Recall the weights R_z: $R_z(i, i; i, i) = 1$, and for $i < j$ we have



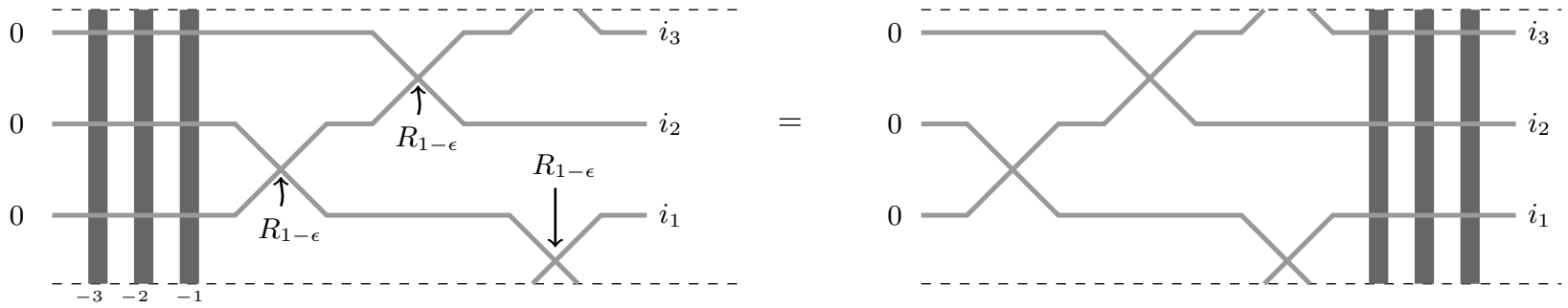
2. Large powers of the matrix $T(\epsilon)$ approximate the time t propagator of mASEP dynamics: $\langle\eta| T^{\lfloor \frac{t}{\epsilon} \rfloor} |\eta'\rangle \to \langle\eta| P^t_{\text{mASEP}} |\eta'\rangle$.



Stationarity via R-matrix commutation

1. We attach our queue model to this Markov transition matrix. The YBE provides the commutation relation

$$\langle \emptyset | \mathfrak{Q}\, T(\epsilon) | \eta \rangle = \langle \emptyset | \mathfrak{Q} | \eta \rangle.$$

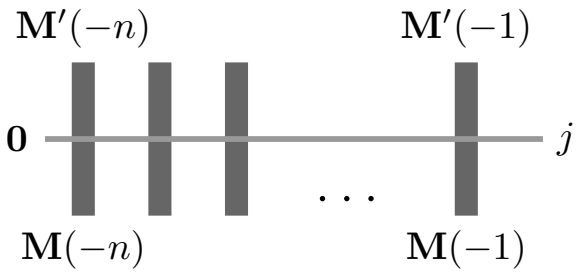


2. This means the covector $\langle \emptyset | \mathfrak{Q}$ is a coeigenvector of eigenvalue 1.
3. In other words, as $\epsilon \to 0$ it provides the stationary measure for the Markov process with generator $\partial_\epsilon T(\epsilon)|_{\epsilon=0}$.
4. *This is the proof of stationarity for the mASEP.* Note: Different specializations of parameters give different multiline queues. (E.g. the *alternative queue* of [Mar20, Section 7].)

Matrix product states from vertex models

1. To view our partition functions as matrix product states, we may define X_j (acting on a tensor product $V_{-n} \otimes \cdots \otimes V_{-1}$) by

$X_j(x)[\mathbf{M}, \mathbf{M}'] =$



2. The normalized queue partition functions are given by

$$\mathrm{Prob}(\eta) = \frac{1}{Z}\mathrm{Tr}\left(\prod_{i=1}^{N} X_{\eta_i}\right).$$

3. In the MPA, one must find matrices $\hat{X}$ such that

$$\sum_{i,i'} X_i X_{i'} (M_{\mathrm{loc}})_{ii',jj'} = X_j \hat{X}_{j'} - \hat{X}_j X_{j'}.$$

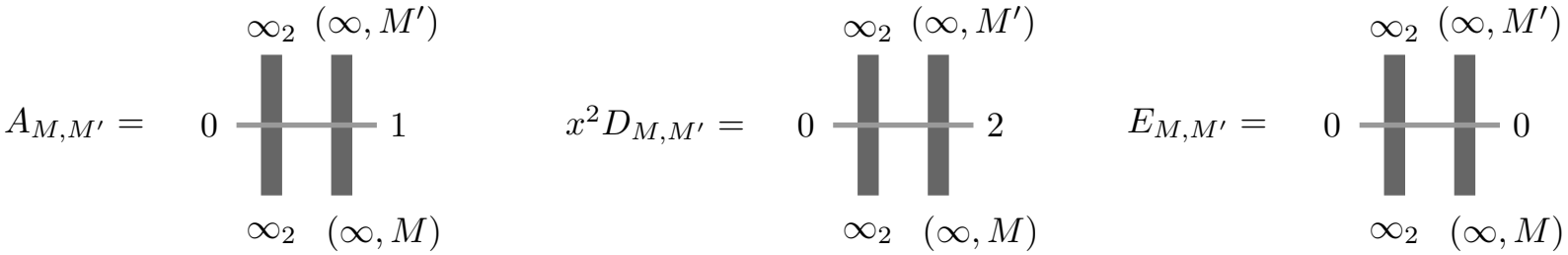
We can use the Yang–Baxter equation to construct these.

A,D,E from vertex weights

1. Usually, the building blocks of matrices X_j are matrices A, D, E satisfying the *quadratic algebra* relations

$$AD - qDA = EA - qAE = (1-q)A, \quad ED - qDE = (1-q)(E+D).$$

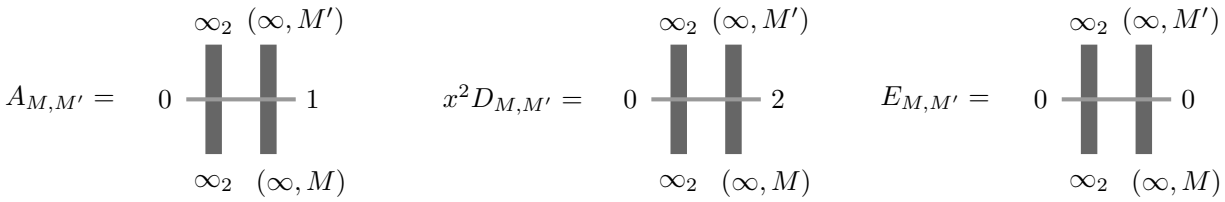
2. In addition, these usually coincide with two-type ($n = 2$) X matrices ($A = X_1, D = X_2, E = X_0$):



3. But our vertex weights contain arbitrary parameters s, x.

A,D,E from vertex weights

1. We define A, D, E via the vertex model:



which leads to

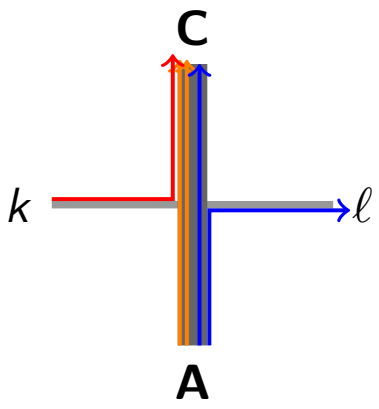
$$A := x \begin{pmatrix} 1 & s & 0 & \dots \\ 0 & q & qs & \dots \\ 0 & 0 & q^2 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \qquad D := x^{-1} \begin{pmatrix} (x-s) & 0 & 0 & \dots \\ (1-q) & (x-sq) & 0 & \dots \\ 0 & (1-q^2) & (x-sq^2) & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix},$$
$$E := \begin{pmatrix} 1 & x & 0 & \dots \\ 0 & 1 & x & \dots \\ 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{3}$$

2. These matrices solve (with two free parameters) the quadratic algebra relations.

Back to $U_q(\hat{sl}_n)$ vertex models

Fused weights

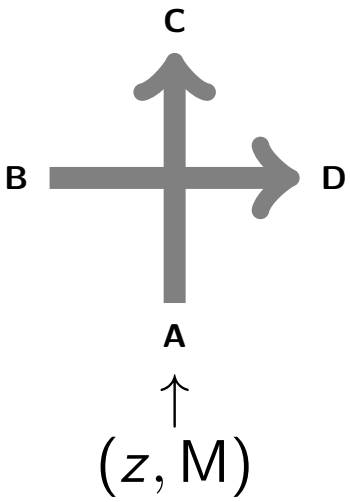
1. Recall: Weights $L_{s,x}$ only allow at most one path per horizontal edge.



2. By a procedure called *fusion* these can be generalized to weights which allow multiple arrows on horizontal edges, as well.

Fused weights

1. The most general vertex weights are

$(y, \mathrm{L}) \rightarrow$ 

$= W_{x,\mathrm{L},\mathrm{M}}(\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D})$

$$:= \mathbf{1}_{\mathbf{A}+\mathbf{B}=\mathbf{C}+\mathbf{D}} \cdot x^{|\mathbf{D}-\mathbf{B}|}(q^{\mathrm{L}})^{|\mathbf{A}|}(q^{\mathrm{M}})^{-|\mathbf{D}|}$$
$$\times \sum\nolimits_{\mathbf{P}} \Phi(\mathbf{C}-\mathbf{P}, \mathbf{C}+\mathbf{D}-\mathbf{P}; q^{\mathrm{L}-\mathrm{M}}x, q^{-\mathrm{M}}x)\Phi(\mathbf{P}, \mathbf{B}; q^{-\mathrm{L}}/x, q^{-\mathrm{L}}),$$

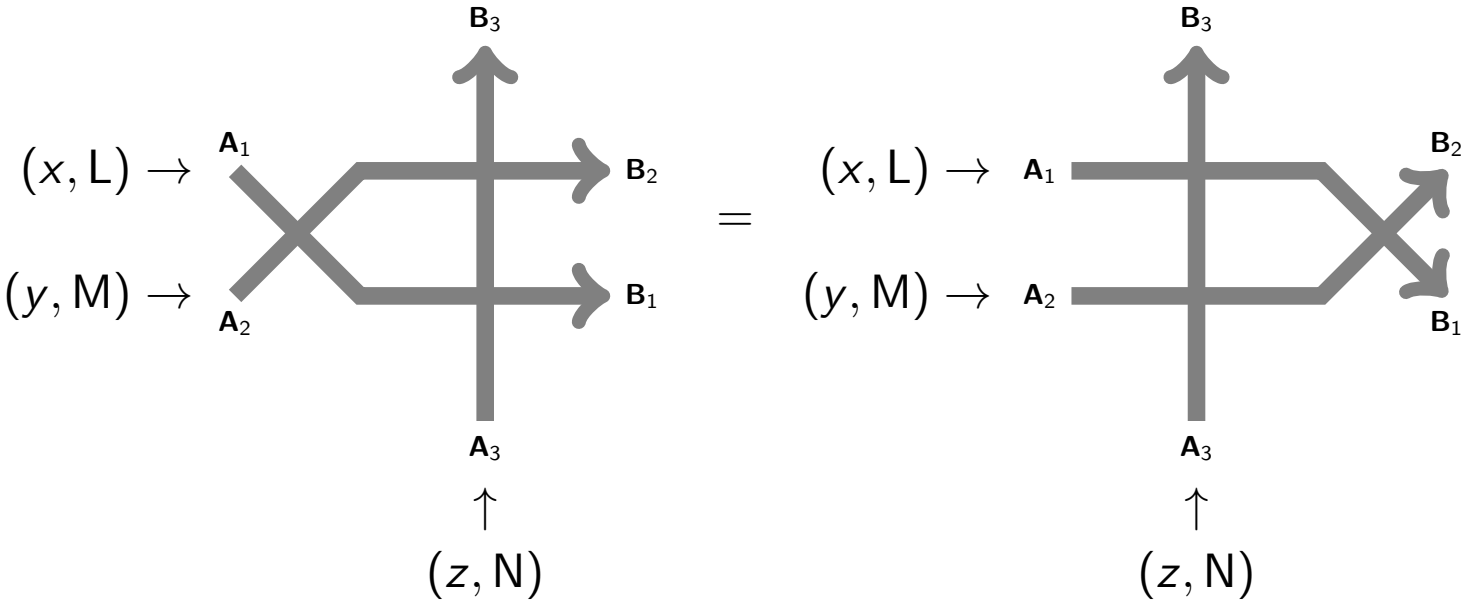
(The sum is over $\mathbf{P} \in \mathbb{Z}_{\geq 0}$ with $P_i \leq \min(B_i, C_i)$, and $x = y/z$.)

2. These weights are also *stochastic*:

$$\sum_{\mathbf{C},\mathbf{D}} W_{x,\mathrm{L},\mathrm{M}}(\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}) = 1.$$

Fused Yang–Baxter

1. The most general Yang–Baxter equation, using weight $W_{x/y,\mathrm{L},\mathrm{M}}$ at crossings of lines with spectral parameters and spin parameters (x, L) and (y, M), respectively, is given by



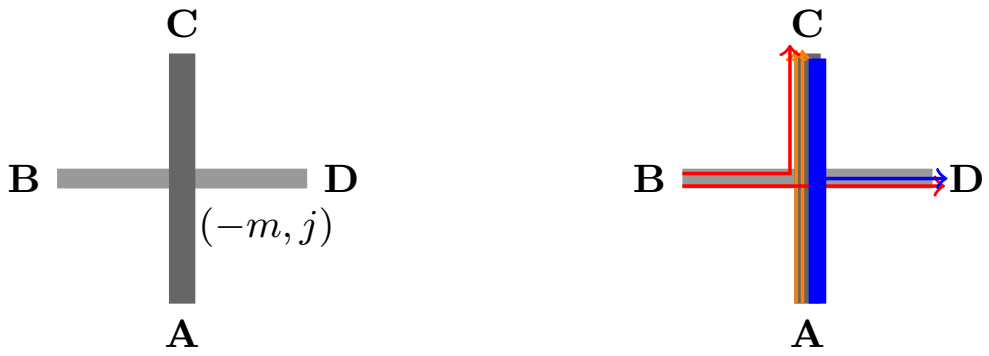
(4)

Queue vertex model and stationary measures

Queue-specialized vertex weights

1. Let $m \in \{1, \ldots, n\}$.
2. We want to define a vertex weight $\mathbb{W}^{(-m)}_{s_1,s_2,u}(\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D})$, on local configurations satisfying $A_m = C_m = +\infty$.
3. Other than $A_m = C_m = +\infty$, the other components of $\mathbf{A} = (A_1, \ldots, A_n), \mathbf{C} = (C_1, \ldots, C_n)$, and each component of $\mathbf{B}, \mathbf{D} \in \mathbb{Z}^n_{\geq 0}$ is a nonnegative integer.
4. Define the *queue specialized vertex weight*

$$\mathbb{W}^{(-m)}_{s_1,s_2,u}(\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}) := \lim_{A_m, C_m \to +\infty} W_{s_1 s_2^{-1} u, \mathrm{L}, \mathrm{M}}(\mathbf{A}, \mathbf{B}; \mathbf{C}, \mathbf{D}). \quad (5)$$



Queue-specialized vertex weights

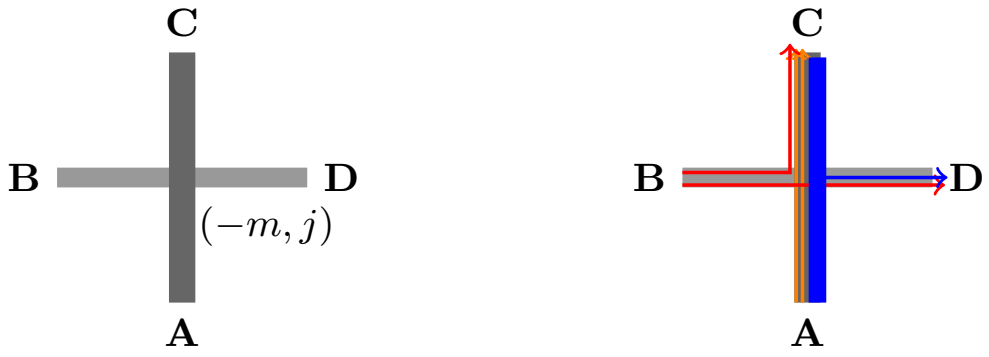
1. Define *queue specialized vertex weights*

$$\mathbb{W}^{(-m)}_{s_1,s_2,u}(\mathbf{A},\mathbf{B};\mathbf{C},\mathbf{D}) := \lim_{A_m,C_m\to+\infty} W_{s_1 s_2^{-1} u,\mathrm{L},\mathrm{M}}(\mathbf{A},\mathbf{B};\mathbf{C},\mathbf{D}). \quad (6)$$

Lemma

The limit (6) *exists and is explicit. Furthermore,* $\mathbb{W}^{(-m)}_{s_1,s_2,u}$ *are stochastic.*

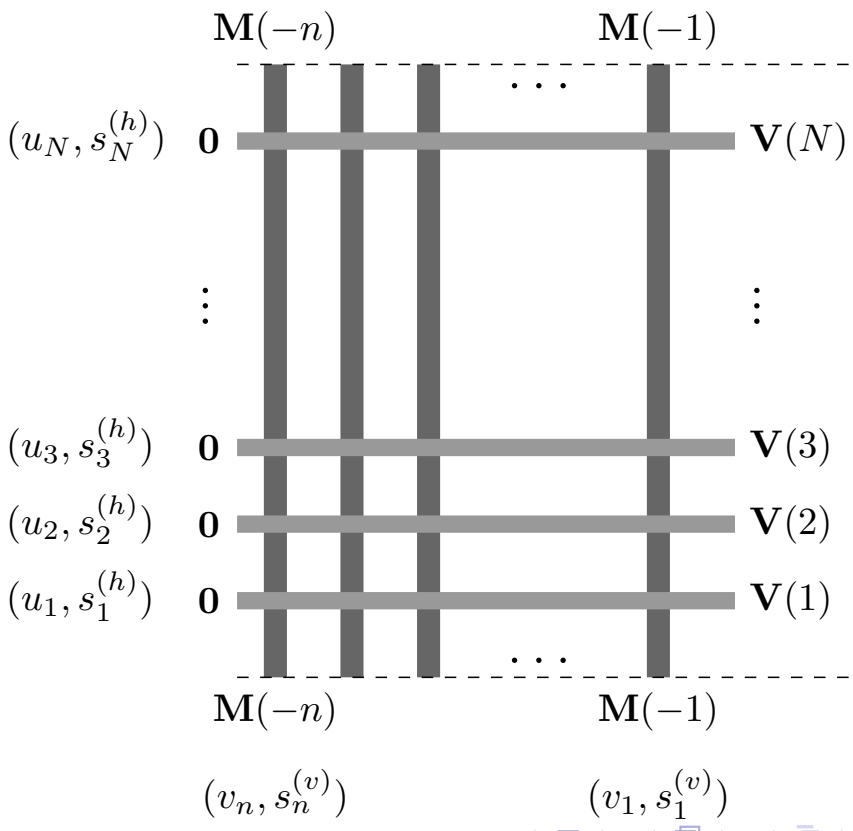
2. Note that if we think of $A_m = C_m = \infty$, we no longer have arrow preservation!



Queue vertex model: parameters and boundary data

1. Choose exit data $\mathbf{V}$, $\mathbf{V}(j) \in \mathbb{Z}^n_{\geq 0}$ encoding a state of the interacting particle system of interest.
2. The *queue vertex model* has the *queue specialized vertex weight* $\mathbb{W}^{(-m)}_{s_j^{(h)}, s_m^{(v)}, u_j/v_m}$ at vertex $(-m, j)$. $\langle \emptyset | \mathfrak{Q} | \mathbf{V} \rangle :=$ Partition function.

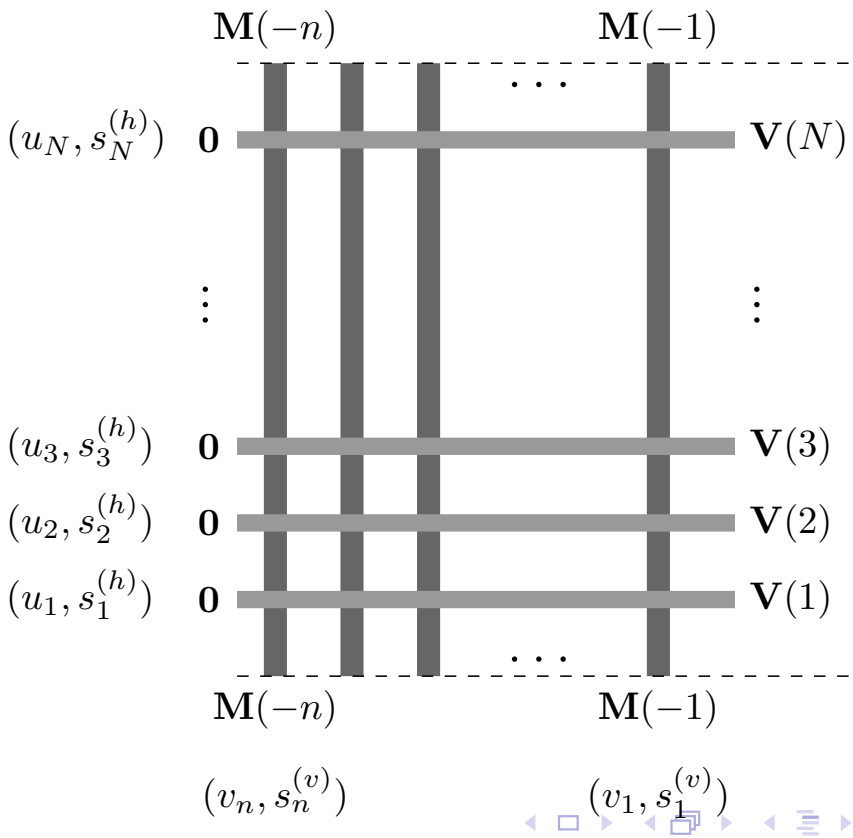
$\sum$



Queue vertex model: sum over windings

1. Each vertical edge of column $-m$ has $\mathbf{M}(-m)_m = +\infty$ color m paths, and $\mathbf{M}(-m)_i = 0$ color i paths for $i < m$. Paths of color m may exit from column $-m$.
2. For each m, we sum over windings of colors $m+1, \ldots, n$ in column $-m$.

$$\sum$$



Stationarity on the ring

Theorem (Aggarwal, N., Petrov)

For each model (mASEP, q-Boson, or q-PushTASEP), by choosing appropriate parameters $u_1, \ldots, u_N$, $s_1^{(h)}, \ldots, s_N^{(h)}$, *we have*

$$\mathrm{Prob}_{N_1,\ldots,N_n}(\mathbf{V}) = \frac{1}{Z}\langle \emptyset | \mathfrak{Q} | \mathbf{V} \rangle.$$

Further degenerating leftover parameters, we are able to match to

1. known mASEP multiline queues and matrix product states [Mar20] [PEM09].
2. a known q-Boson multiline queue stationary measure [AMM22].

We also obtain new stationary measures for q-PushTASEP.

Stationary measures in $\mathbb{Z}$ and for colored stochastic vertex models in the quadrant $\mathbb{Z}^2_{\geq 0}$

Stationary measures on $\mathbb{Z}$

1. For each particle system, extremal translation invariant stationary measures on $\mathbb{Z}$ are expected (or known) to be parameterized by local particle densities $\rho = (\rho_1, \ldots, \rho_n)$ for each type.

Stationary measures on $\mathbb{Z}$

1. For each particle system, extremal translation invariant stationary measures on $\mathbb{Z}$ are expected (or known) to be parameterized by local particle densities $\rho = (\rho_1, \ldots, \rho_n)$ for each type.
2. More precisely, define the local density of particle color i, $i = 1, \ldots, n$, as

$$\rho_i = \mathbb{E}\left[\# \text{ color } i \text{ particles at site } 0\right].$$

Stationary measures on $\mathbb{Z}$

1. For each particle system, extremal translation invariant stationary measures on $\mathbb{Z}$ are expected (or known) to be parameterized by local particle densities $\rho = (\rho_1, \ldots, \rho_n)$ for each type.
2. More precisely, define the local density of particle color i, $i = 1, \ldots, n$, as

$$\rho_i = \mathbb{E}\left[\# \text{ color } i \text{ particles at site } 0\right].$$

3. For each of these systems, we construct stationary measures on $\mathbb{Z}$ using a stochastic vertex model. Densities $\rho = (\rho_1, \ldots, \rho_n)$ are controlled by the leftover parameters in our vertex model.

Stationary measures on $\mathbb{Z}$

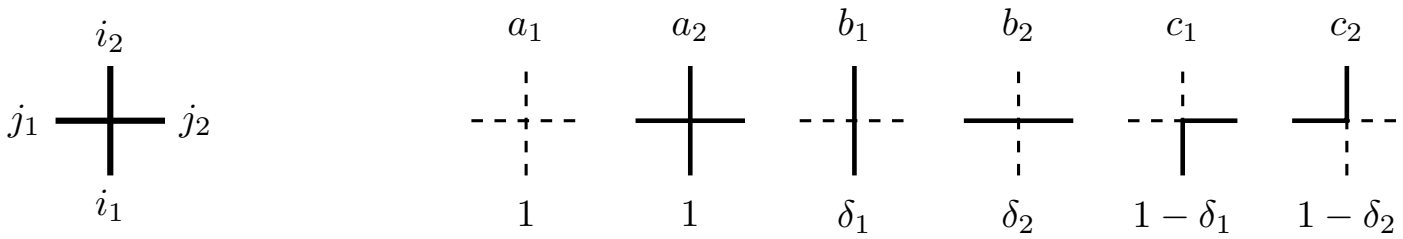
1. For each particle system, extremal translation invariant stationary measures on $\mathbb{Z}$ are expected (or known) to be parameterized by local particle densities $\rho = (\rho_1, \ldots, \rho_n)$ for each type.
2. More precisely, define the local density of particle color i, $i = 1, \ldots, n$, as

$$\rho_i = \mathbb{E}\left[\# \text{ color } i \text{ particles at site } 0\right].$$

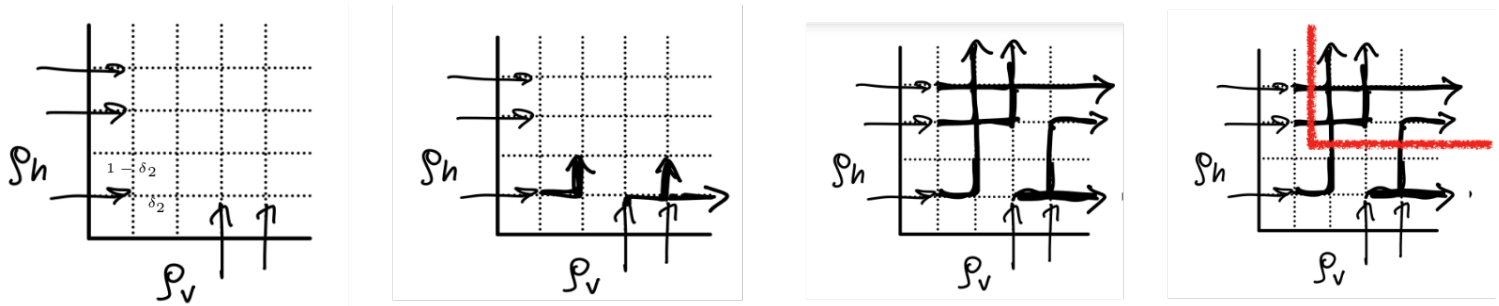
3. For each of these systems, we construct stationary measures on $\mathbb{Z}$ using a stochastic vertex model. Densities $\rho = (\rho_1, \ldots, \rho_n)$ are controlled by the leftover parameters in our vertex model.
4. Instead of a bulk limit on the ring, we construct a *stationary configuration in the boundary of the quadrant*.

Stationary stochastic six-vertex model in a quadrant: single color case

1. Recall *stochastic* R-matrix weights: $\sum_{i_2,j_2} w(i_1,j_2;i_2,j_2) = 1$.

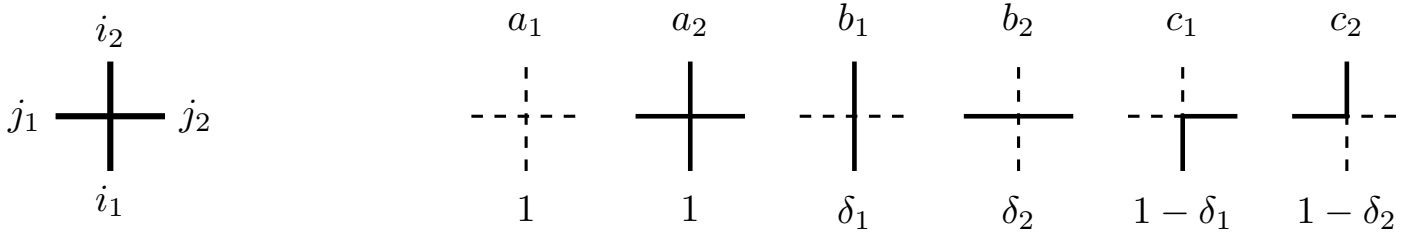


2. Sample an arrow entering at $(j,0)$, $j \geq 0$, with probability $\rho_v = \rho$ and entering at $(0,j)$, $j \geq 0$, with probability $\rho_h = \varphi(\rho_v)$.

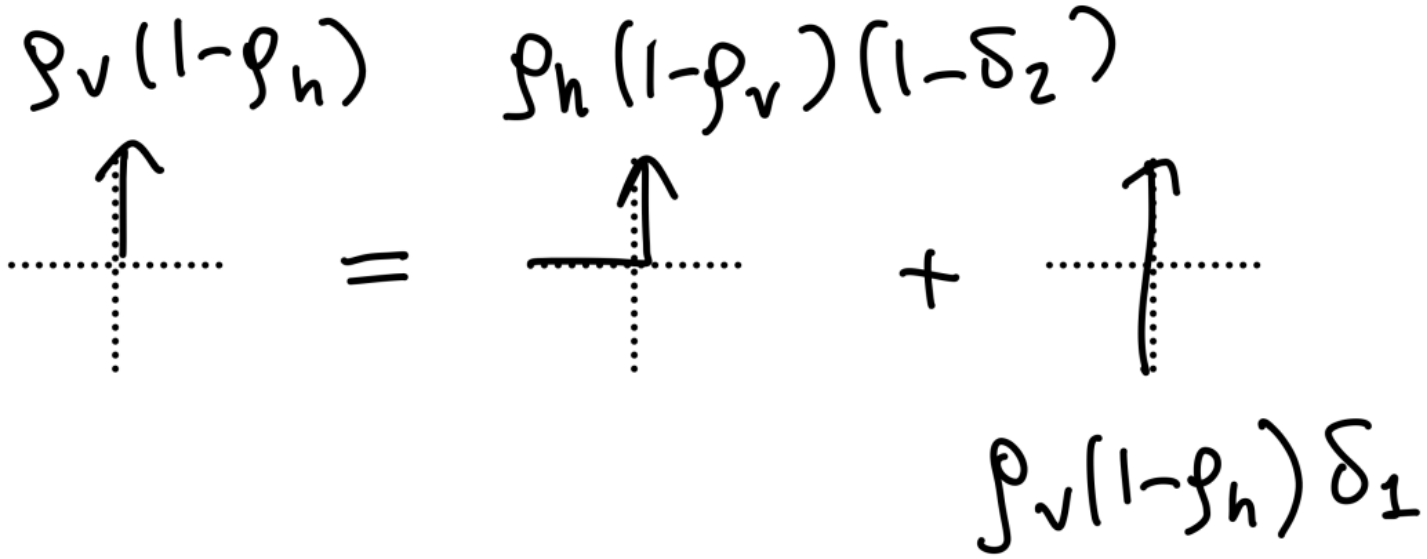


Stationary stochastic six-vertex model in a quadrant from Burke's property

1. Recall *stochastic* R-matrix weights: $\sum_{i_2,j_2} w(i_1,j_2;i_2,j_2) = 1$.

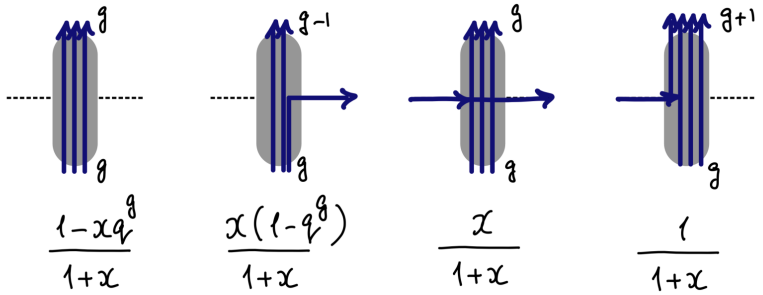


2. One proof of stationarity is a direct check, e.g. if $\rho_h = \varphi(\rho_v)$, then we obtain the *Burke property*

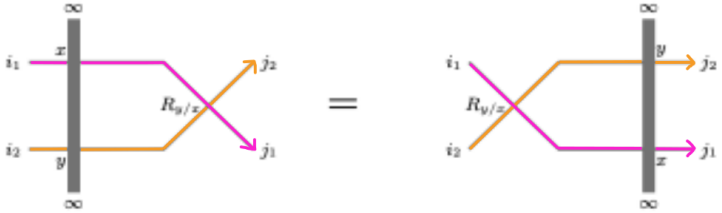


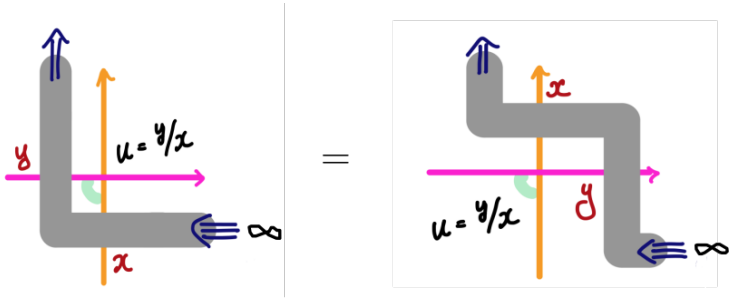
Burke's property from YBE

1. We recall the YBE with the *stochastic* R-matrix weights and *stochastic* queue vertex weights $\mathbb{W}^{(-1)}_{s,x}$, obtained by sending $g \to \infty$ below.



2. The Yang–Baxter equation provides the Burke property: Below, if vertical arrows have probability ρ_v, then $\rho_h = \varphi(\rho_v)$ and all arrows are independent.

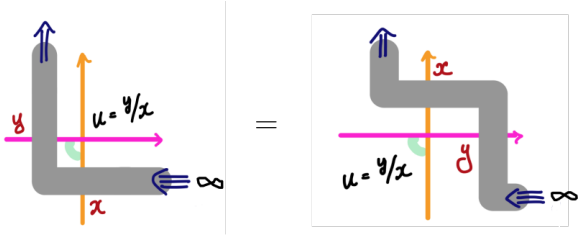


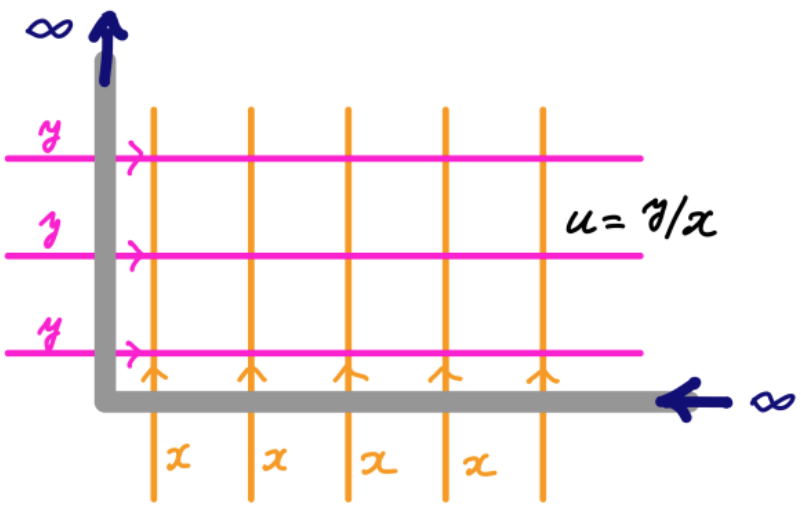


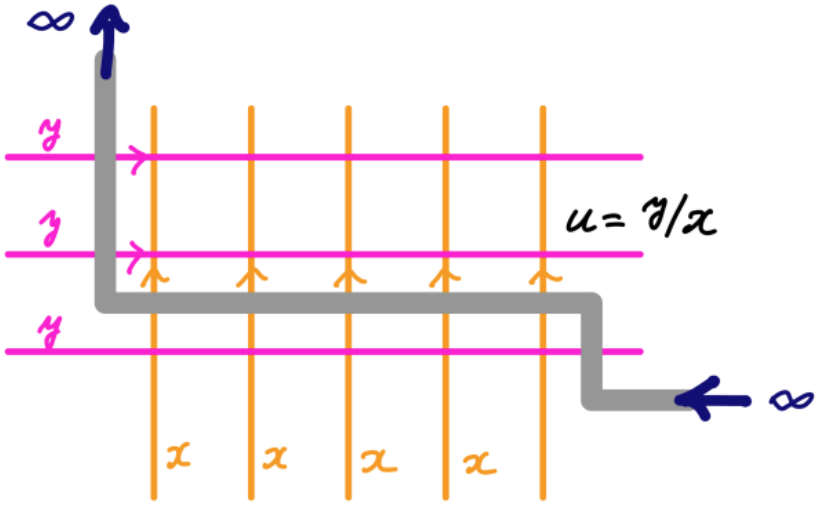
YBE with queue weights.

Stationarity from YBE

1. Infinitely many arrows travel along the thick line.
2. Repeated applications of the YBE give a direct proof of stationarity.

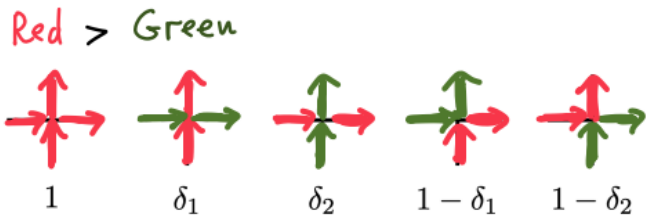


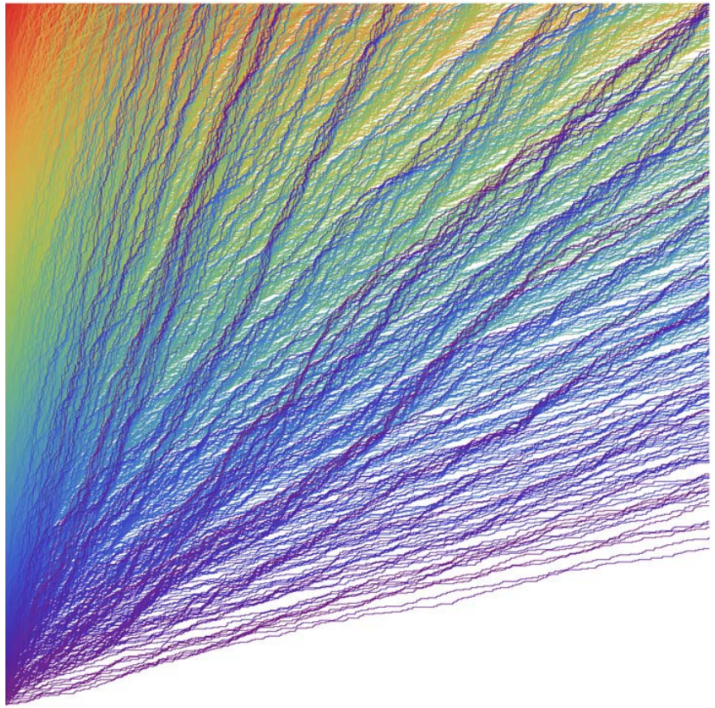




Stochastic six-vertex model in a quadrant: multi-colored case

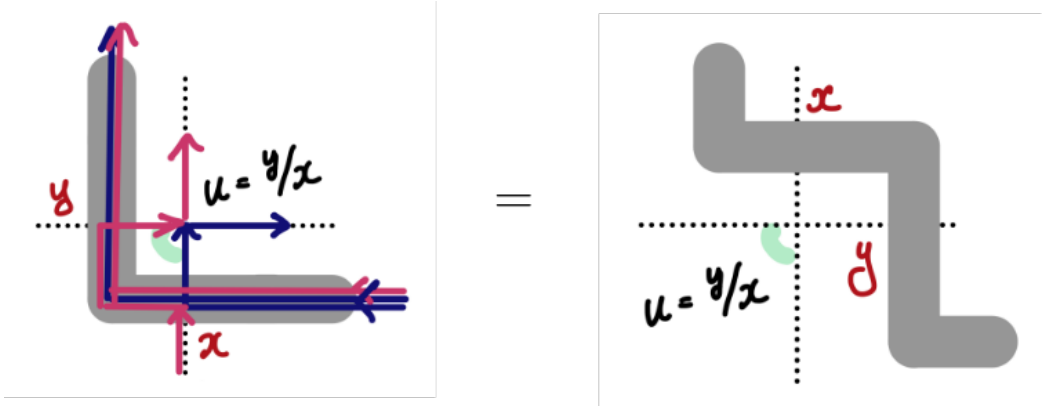
Recall *colored stochastic* R-matrix weights.





A colored analog of the Burke property

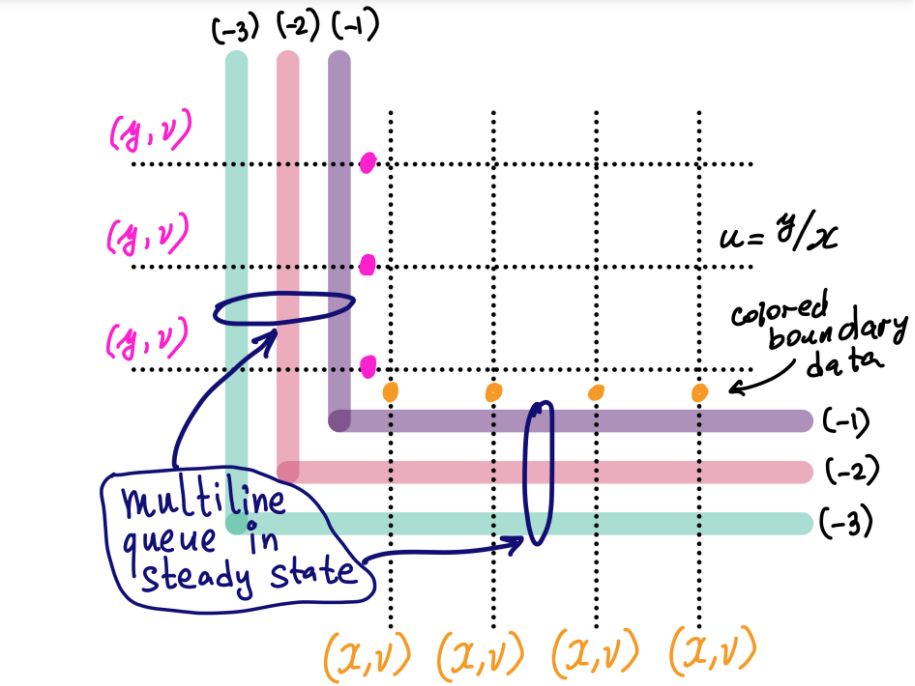
1. The Yang–Baxter equation for the colored R matrix tells us:



2. We obtain the *queue weights* from before by sending the number of type m paths $\to +\infty$.
3. We may choose parameters so *all vertex weights are nonnegative*, so each vertex is a stochastic evolution.

Stationary stochastic six-vertex model in a quadrant: multi-colored case

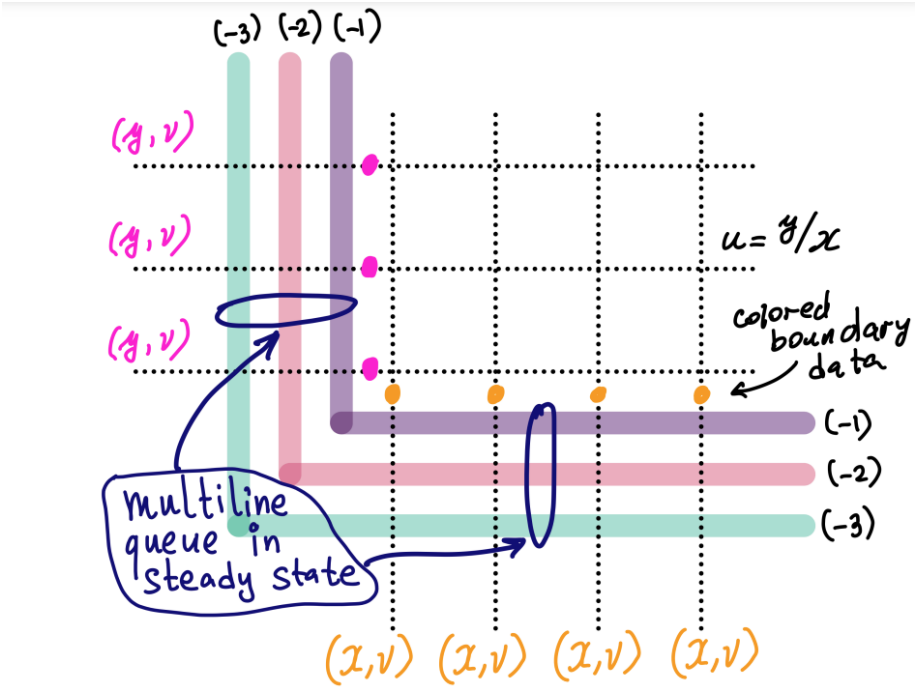
1. In the colored case, we may try to construct stationary measures states similarly.
2. How should we set boundary conditions?
3. The arrows along thick horizontal lines should be sampled from their *steady state*.



Stationary stochastic six-vertex model in a quadrant: multi-colored case

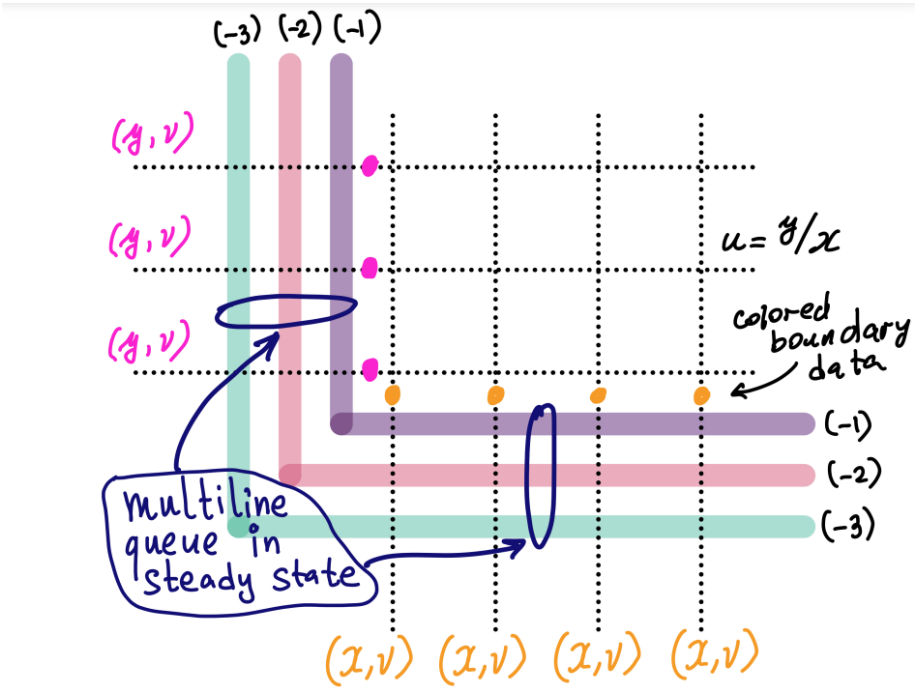
Theorem (Aggarwal, N., Petrov)

Let the process evolving along thick lines be stationary, and let incoming spin 1/2 edges be empty. Then, the boundary data sampled as shown below is stationary for the quarter plane stochastic six-vertex model.



A stationary quarter plane vertex model (spin 1/2 case)

1. This is analogous to a *KPZ pure state* in the single color case.
2. Here $\mathbb{E}[\mathbf{1}\{d(i)=m\}] = \rho_m^{(v)} = \frac{\alpha_m q^{-1/2}}{1+\alpha_m q^{-1/2}} - \frac{\alpha_{m+1} q^{-1/2}}{1+\alpha_{m+1} q^{-1/2}}$, and $\mathbb{E}[\mathbf{1}\{\eta(i)=m\}] = \rho_m^{(h)} = \frac{\rho_m^{(v)} u}{\left(1+(u-1)\rho_{[m,n]}^{(v)}\right)\left(1+(u-1)\rho_{[m+1,n]}^{(v)}\right)}$ are determined by the parameters $\{\alpha_i\}_{i=1}^m$ (and of the actual model, u).



Future directions

Future directions

1. Construct stationary measures of half space and segment models. More complicated versions of this construction may work. Might provide a chance to take asymptotics in new regimes (speculative).
2. Study a version of *stationary horizon* on the circle, or prove convergence of a new model to the stationary horizon. See recent work of Corwin, Gu, and Sorensen on stationary coupled solutions of the KPZ equation on the circle https://arxiv.org/pdf/2409.03613.

References I

[AMM22] Arvind Ayyer, Olya Mandelshtam, and James B Martin.

Modified macdonald polynomials and the multispecies zero range process: Ii.
arXiv preprint arXiv:2209.09859, 2022.

[Baz85] V. Bazhanov.
Trigonometric solutions of triangle equations and classical Lie algebras.
Phys. Lett. B, 159:321?324, 1985.

[BW18] A. Borodin and M. Wheeler.
Coloured stochastic vertex models and their spectral theory.
arXiv preprint, 2018.
arXiv:1808.01866 [math.PR].

References II

[CdGW15] L. Cantini, J. de Gier, and M. Wheeler.
Matrix product formula for macdonald polynomials.
Journal of Physics A: Mathematical and Theoretical, 48(38):384001, 2015.
arXiv:1505.00287 [math-ph].

[CK23] S. Corteel and D. Keating.
In preparation.
2023.

[CMW18] S. Corteel, O. Mandelshtam, and L. Williams.
From multiline queues to macdonald polynomials via the exclusion process.
arXiv preprint, 2018.
arXiv:1811.01024 [math.CO].

References III

[FM07] Pablo A. Ferrari and James B. Martin.
Stationary distributions of multi-type totally asymmetric exclusion processes.
Ann. Probab., 35(3):807–832, 2007.

[Jim86] M. Jimbo.
Quantum R matrix for the generalized Toda system.
Commun. Math. Phys., 102:537–547, 1986.

[Mar20] J. Martin.
Stationary distributions of the multi-type asep.
Electronic Journal of Probability, 25(43):1–41, 2020.
arXiv:1810.10650 [math.PR].

References IV

[PEM09] S Prolhac, M R Evans, and K Mallick.
The matrix product solution of the multispecies partially asymmetric exclusion process.
Journal of Physics A: Mathematical and Theoretical, 42(16):165004, mar 2009.